

Effect of Partial Ground and Partial Ceiling on Propeller Performance

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Finite extent flat obstacles behind a propeller (“partial ground”) and ahead of the propeller (“partial ceiling”) were examined experimentally for a small-scale fixed-pitch propeller in hover. Cases included circular and annular plates, which can be summed up or superimposed. Near-surface smoke and TiO_2 flow visualization show a projected time-averaged streamline on the ground plate, dividing a swirling flow inboard and a radial flow outboard. The flow pattern was juxtaposed with pressure data on the ground plate. Changes in propeller thrust and power due to ground/ceiling effect were measured by a force balance and correlated with flow visualization. For a plate of diameter equal to that of the propeller, ground effect on thrust and power was comparable to that of an “infinite” plate, while the flowfield also resembles that of the infinite plate. For a plate of less than half of the propeller diameter, ground or ceiling effect was found to be negligible. Thrust and power effects for circular and respective annular plates were superimposed to determine whether the increase in thrust or decrease in power required is itself additive to the respective effects seen with the constituent obstacles. For ceiling effect, this holds but less so for ground effect. A curve fit to thrust and power data resulted in a predictive model accurate to within 6%.

Nomenclature

B	=	ratio of circular ground-plate outer diameter to propeller diameter; d_p/D
C_P	=	propeller power coefficient; $P/(\rho n^2 D^5)$
C_Q	=	propeller torque coefficient; $Q/(\rho n^2 D^5)$
C_T	=	propeller thrust coefficient; $T/(\rho n^2 D^4)$
D	=	propeller diameter, m
D_p	=	downforce on ground plate, N
d_o	=	ground-plate opening diameter, m
d_p	=	ground-plate diameter, m
F_{net}	=	net force, N
h	=	distance to ground plate, m
n	=	propeller rotational speed per second
O	=	ratio of annulus inner diameter to propeller diameter; d_o/D
P	=	power, W
P_{def}	=	normalized power required deficit
p	=	pressure, Pa
Q	=	torque N · m
R	=	propeller radius, m
S	=	propeller disk area, m^2
T	=	thrust, N
γ	=	propeller pitch, m
ρ	=	air density, kg/m^3
σ	=	propeller solidity; $S/(0.25\pi D^2)$

Subscripts

D_{norm}	=	force normalized by plate force
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Emp	=	empirical fit
Mod	=	prediction from model
T_{norm}	=	force normalized by propeller thrust

I. Introduction

GROUND effect, or particularly the effect on rotor/propeller performance, of an impermeable planar surface parallel and in close proximity to the rotor has been studied theoretically [1–3] and experimentally [2–9] for decades. Ground plate offers the following two benefits: an increase in thrust produced and a decrease in power required for a given rotational speed, which combine into propulsive efficiency augmentation [10]. Research on the small-scale unmanned aerial vehicle (UAV) propeller/rotor ground effect has recently garnered more attention, with the emerging use of quadcopters [11–20]. Many novel studies consider multirotor ground effect, which is rare for large rotor applications. He and Leang [11] consider multicopter ground-effect thrust augmentation, finding up to a 5% difference in thrust generation within a rotor-to-rotor distance of $0.25D$. Dekker et al. [12] studied the ground effect on a side-by-side rotor using three-dimensional particle image velocimetry. A fountain flow is observed between two rotors within a rotor-to-rotor distance of $2.0D$; with the decrement of the propeller-to-ground distance, a reingestion/recirculation fountain flow is observed, resulting in a significant reduction in rotor thrust generation in ground effect (IGE). When placing the rotor within $0.25D$ from each other, a merged wake between the rotor occurs, which brings benefit on the rotor thrust generation.

Small UAVs are often operated in a confined space, with the possibility of encountering both a “ground plate” at the outflow side of the propeller and a “ceiling plate” at the inflow side. Several recent works have considered the ceiling effect [13–17]. Just as with the classical ground effect, thrust augmentation is observed in the ceiling effect, although the classical ground-effect prediction model cannot be applied to the ceiling effect due to the significant differences that were observed in the flowfield [15,17]. Several modern drone concepts take advantage of the ceiling-effect thrust benefit [18–20] and operate constantly in the ceiling effect to increase range.

A large flat surface is typically assumed in ground-effect studies, while “partial ground effect,” where the rotor encounters a surface that is smaller than the rotor area, is less common. For large helicopter rotors, partial ground effect can be encountered when landing on a

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ship deck. Xin et al. [21] modeled the partial ground effect as a function of rotor wake momentum and ground area under the rotor disk. Sanchez-Cuevas et al. [22] examined the stability of quadcopter flight over obstacles, where partial ground effect will be experienced by the rotor. A significant impact on quadcopter pitch angle is found when flying less than one propeller diameter above the ground. Operation of drones over uneven surfaces and small obstacles is common for small-scale UAV applications. Sanchez-Cuevas et al. [13] also consider drones approaching an obstacle, such as piping or blocks, finding differences between partial ground effect and partial ceiling effect.

Besides considering the role of proximity of finite size obstacles, it is useful to assess how large the ground/ceiling plate should be, relative to propeller diameter, to be considered “infinite.” The limiting cases are a round plate that is sufficiently large to be infinite, or in the opposite, one along the axis of the propeller that is sufficiently small as to be indistinguishable from out of ground effect (OGE). In our prior work [10], we considered that a ratio of ground-plate standoff distance h to propeller diameter D is as little as 0.05. The propeller orientation is then reversed, obtaining a ceiling-effect condition. The ground-plate blockage ratio B , which is the ratio of plate characteristic diameter d_p to propeller diameter, was ~ 4 . Multiple APC thin electric propellers [23] were tested, with different pitch-to-diameter ratios and solidity. Normalized power required at constant thrust, P_{IGE}/P_{OGE} , was used as a metric to capture both thrust augmentation and reduction in power required. A model, applicable to propeller pitch γ -to-diameter D ratios of $0 < (\gamma/D) < 0.636$, was proposed.

Near-surface flow visualizations via a dense array of tufts [10], laser-illuminated smoke visualization, and titanium dioxide oil mixture surface images are compared in Fig. 1, for the APC 17 $\times$ 17E (0.43 m diameter $\times$ 0.18 m pitch) propeller at $h/D = 0.2$ and nominally infinite ground plate ($B = \infty$). A projected “dividing streamline,” approximately circular with a diameter on the order of half of the propeller diameter, is observed from the tuft and oil flow visualization. Exterior to this circular boundary, the projected streamlines appear to be radially outward. Interior to the circular boundary, the projected flow can be described as tangential and swirling, without a discernible radial component. This phenomenon is also observed in smoke flow visualization at $h/D \leq 0.3$. Based on the force measurement and flow visualization result, it is hypothesized that the thrust augmentation and power reduction in ground effect relate to trends in the tangential and radial flow regions.

The idea of a circular and annular ground plate then emerges. The former has the center plate region preserved, while the radial flow region is removed, to study the effect of the tangential flow on propeller performance. The latter has the center region of the plate removed, eliminating the tangential flow region, to study the effect of radial flow on propeller performance. These conditions are then called partial ground/ceiling effect. As most of the aforementioned literature as well as our previous study consider a ground-plate size much larger than the propeller diameter, it is useful to understand the effect of ground-plate size on the overall propeller performance and to what extent can the ground plate be called “infinite ground.”

II. Experimental Setup

Experiments were conducted on a thrust stand at the University of Dayton. Propellers were driven by an E-Flite Power 60–400 kV brushless outrunner electric motor, driven by a PSW 30-108 constant-voltage power supply. A photodiode was used to measure the propeller rotational speed (n). A schematic of the setup is given in Fig. 2.

A. Thrust and Torque Measurements

The propeller motor was attached to an ATI Industrial Automation Mini40 six-component force balance [24]. Thrust is the axial force along the z axis of the sensor, and torque is a rotational moment of the z axis. The balance bolts to an aluminum extrusion frame secured to the laboratory floor. The balance has a maximum of 120 N z -axis force with 0.02 N resolution and maximum Z -axis torque of 2 N $\cdot$ m with a 0.00025 N $\cdot$ m resolution. The encountered thrust and torque range were 7–40 N and 0.15–1.2 N $\cdot$ m, respectively. For all circular plates, an ATI Gamma force balance [25] (Z -axis maximum 200 N; resolution 0.025 N) was used to measure the normal force on circular ground plates. For annular plates, direct force measurement was not possible. To achieve negative h/D for ceiling effect, the propeller is reversely attached to the motor, and the motor rotation direction is also reversed. Because the test articles are fixed-pitch propellers developed for UAVs or model airplanes, the measured forces and torques are normalized using conventions for fixed-wing aircraft, rather than for rotorcraft.

The data sampling rate was 100 Hz for a 15 s duration at each data point, with two runs to improve data repeatability for each data point (value of rpm, propeller pitch/diameter, and h/D). To minimize sensor drift, tare values with the propeller powered off were taken before and after each test. The average of the two tares was subtracted from the respective thrust and torque readings. If the difference between the before and after tare values is greater than 0.1 N of force on any axis, the experimental data are then rejected, and a repeat run is performed. A finite impulse response band-stop filter [26] was applied to the data to filter out the vibration frequency caused by the propeller rotation, which follows the propeller rotational speed, and to the natural frequency of the aluminum frame (~ 60 Hz), which is obtained from the tap testing with propeller powered off.

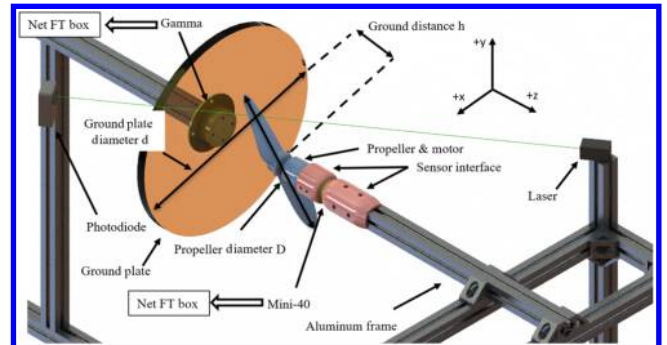


Fig. 2 Schematic (isometric view) of thrust stand and ground plate.

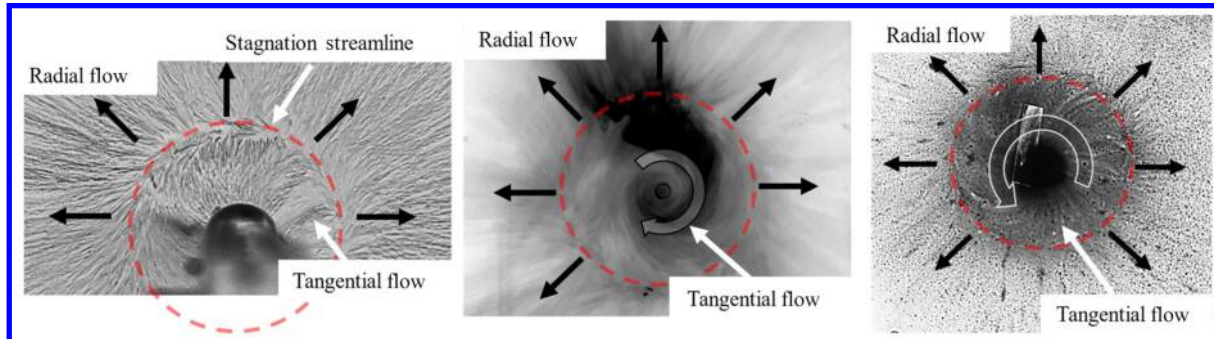


Fig. 1 Tuft (left), smoke (middle), and oil (right) visualization of near-surface flowfield, $h/D = 0.2$.

To attenuate flowfield interference due to the thrust stand itself, which is to say blockage unrelated to the ground effect, the propeller disk was placed at greater than 1 diameter away from any transverse members of the aluminum frame of the thrust stand. The 3-D-printed plastic streamline cowling covering the sensor is <2% of the propeller-disk area. To minimize the wall effect from the laboratory floor, the tip of the propeller is mounted at least two propeller diameters above the laboratory floor.

The APC 17 × 7 “thin electric” propeller was selected for most of the test campaign because of its small ratio of pitch to diameter. Our previous study on infinite ground plates [10] showed that ground-effect benefits are larger for propellers of a lower pitch-to-diameter ratio, γ/D . The 17 × 7 has $\gamma/D = 0.412$ and solidity $\sigma = 0.0846$. Seven different rotational speeds between 50 and 100 rotations per second (RPS) were tested for each h/D , resulting in blade Reynolds numbers of 100,000–200,000, calculated at 75% blade radius. The APC 17 × 10 and 17 × 12 propellers, with respectively higher γ/D and lower IGE thrust augmentation, were used as additional force and torque test cases for circular ground plates. The APC 14 × 6 propeller, which also has low γ/D (0.429), was used as a complementary case for validation of an empirical performance model.

A common representation of propeller IGE thrust and power performance is the “normalized power” (P_{IGE}/P_{OGE}) at constant thrust. As mentioned previously, the ground effect and the ceiling

effect manifest themselves simultaneously as an increase in thrust and decrement in power required. The two effects can be combined as P_{IGE}/P_{OGE} . To calculate this, the propeller IGE and OGE thrust and torque are measured over a range of propeller rotational speeds between 50 and 100 RPS. For every h/D of interest, a second-order interpolation for thrust vs rpm and a third-order interpolation of torque vs rpm across the rotational speed are created. Then, a thrust level of interest is selected, while the rotational speed at the selected thrust level is found using the thrust interpolation at every h/D . At this particular rotational speed, the IGE power is obtained from the torque interpolation. Results are averaged across OGE thrust levels, that is, across propeller rotational speed values.

B. Ground-Plate Nomenclature

The ratio of the outer diameter of the ground plate to that of the propeller is defined as $B = d_p/D$, while the ratio of the ground-plate inner diameter to the propeller diameter is defined as $O = d_o/D$. Thus, a circular plate with half of the propeller diameter is $B_{0.5}O_{0.0}$, an annulus with the same outer diameter as the propeller and an opening half of the propeller diameter is $B_{1.0}O_{0.5}$, an infinite ground plate with an opening half of the propeller diameter is $B_{\infty}O_{0.5}$, and the usual infinite ground plate becomes $B_{\infty}O_{0.0}$. Table 1 lists the test conditions, and Fig. 3 shows a pictorial description of the naming convention.

Table 1 Ground-plate nomenclature

Infinite ground ($B = \infty; O = 0.0$)	Circular ground ($B \neq 0; O = 0.0$)	Annular ground ($B = 1.0; O \neq 0.0$)	Annular infinite ground ($B = \infty; O \neq 0.0$)
$B_{\infty}O_{0.0}$	$B_{0.5}O_{0.0}; B_{0.625}O_{0.0}; B_{0.75}O_{0.0}; B_{1.0}O_{0.0}$	$B_{1.0}O_{0.5}; B_{1.0}O_{0.625}; B_{1.0}O_{0.75}$	$B_{\infty}O_{0.625}; B_{\infty}O_{0.75}$

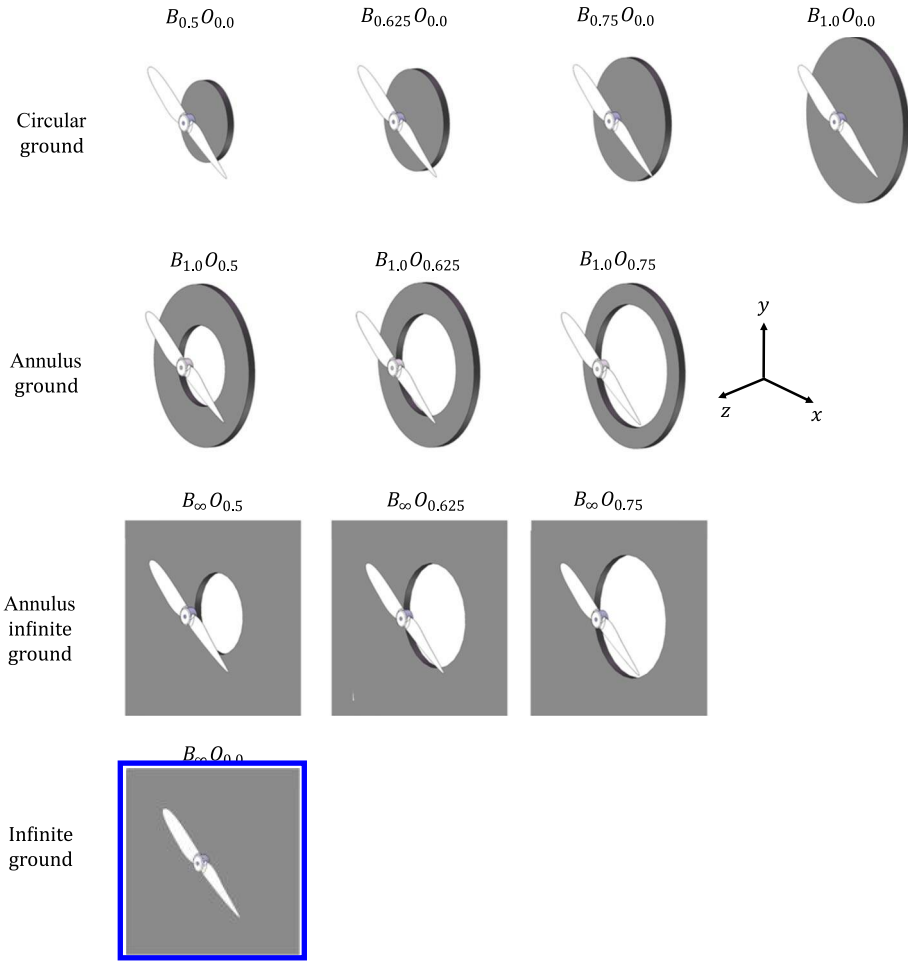


Fig. 3 Ground-plate schematic and naming convention.

C. Flow Visualization

Smoke flow visualization was performed at rpm ~ 900 to improve the image quality while still preserving the flowfield of the higher rpm cases. The smoke wand has a diameter of 0.5 in., placed $0.5D$ upstream of the propeller disk. A 200 mW continuous laser was used as a light source. For the oil surface flow visualization, a mixture of titanium dioxide, mineral oil, and oleic acid is applied on the ground plate coated with black enamel at rpm ~ 5400 . The higher rotational speed improves the relaxation of the oil pattern to the steady state.

D. Unsteady Pressure-Transducer Measurement

Seven Kulite XTEL-10L-190 pressure transducers [27] were placed flush on the circular ground plates and annular ground plates. The radial locations of the transducers are shown in Table 2. All have a calibrated range of ± 69 kPa gauge pressure and an averaged unfiltered measurement uncertainty of 9%. Data were collected at a sampling rate of 5 kHz and averaged over 5 s. Pressure is normalized by the spatial average plate pressure calculated from the thrust measurement $p_d = 4D_p/\pi d_p^2$ for the $B_{1.0}O_{0.0}$ case at each respective h/D . Data were taken in the h/D range from 0.1 to 0.5 in. $0.1h/D$ increments.

III. Results

Normalized power at constant thrust is shown in Fig. 4, for all cases in Table 1, split into circular (Fig. 4a) and annular (Fig. 4b) plates. Five OGE thrust levels, between 15 and 35 N, are selected within the rotational speed range tested, and the resulting average value is shown. The variation among different thrust levels is less than 0.7% on average. Here, as in our earlier results [10], P_{IGE}/P_{OGE} does not quite attain 1.0 at $h/D = 1.5$, although it does reach nearly 1.0 on the ceiling-effect side at $h/D = -1.5$. Based on our previous study [10], P_{IGE}/P_{OGE} attains 1.0 at $h/D > 4$ for ground effect. This is likely caused by the flow recirculation due to the presence of the laboratory floor, which leads to a small increment in the propeller inflow velocity close to the propeller tip. A similar result is also obtained from other studies [11,15]. In the “ground-effect” configuration, the thrust stand is upstream of the propeller inflow, while in the “ceiling effect” the thrust stand is in the propeller wake, which could also contribute to the 3% difference in normalized power at constant thrust as an installation blockage. This is a fixed bias error. Noting this, we focus on $|h/D| < 1.5$.

First considering the circular plates, $B_{1.0}O_{0.0}$ closely follows $B_{\infty}O_{0.0}$ at all h/D . This means that a circular plate whose diameter

Table 2 Pressure-transducer placement

Ground plate	Pressure-transducer radial location (r/R) marked as “x”												
	0.24	0.29	0.35	0.41	0.47	0.53	0.59	0.65	0.71	0.76	0.82	0.88	0.94
$B_{1.0}O_{0.0}$	x	—	x	—	x	—	x	—	x	—	x	—	x
$B_{0.5}O_{0.0}$	x	x	x	x	x	—	—	—	—	—	—	—	—
$B_{0.625}O_{0.0}$	x	x	x	x	x	x	x	—	—	—	—	—	—
$B_{0.75}O_{0.0}$	x	—	x	x	x	x	x	—	x	—	—	—	—
$B_{1.0}O_{0.5}$	—	—	—	—	—	x	x	x	x	x	x	x	—
$B_{1.0}O_{0.625}$	—	—	—	—	—	—	—	x	x	x	x	x	x
$B_{1.0}O_{0.75}$	—	—	—	—	—	—	—	—	—	x	x	x	—

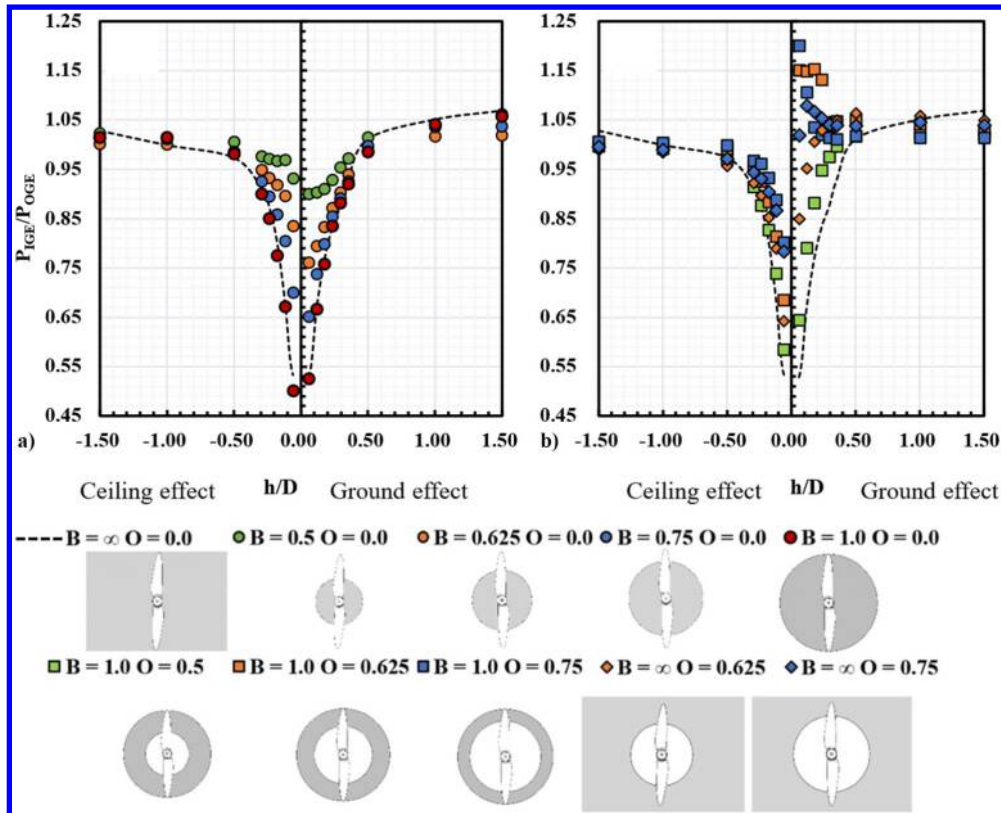


Fig. 4 Normalized power required for constant thrust: a) circular and b) annular plates.

is the same as that of the propeller appears to be already large enough to serve as infinite ground. As plate diameter decreases, the reduction in P_{IGE}/P_{OGE} attenuates monotonically until finally, for $B_{0.5}O_{0.0}$, the reduction of P_{IGE}/P_{OGE} becomes negligible. That is, by the metric of P_{IGE}/P_{OGE} , $B_{0.5}O_{0.0}$ approaches OGE. For the ceiling effect, where $h/D < 0$, the results for all cases follow the trend for traditional ground effect in [10], where P_{IGE}/P_{OGE} decreases with the decrease in h/D .

Turning to the annular plates and in particular to $h/D < 0$, $B_{1.0}O_{0.5}$ has a drop in P_{IGE}/P_{OGE} of some 80% of that of the reference infinite ground, $B_{\infty}O_{0.0}$, $B_{1.0}O_{0.75}$ and $B_{\infty}O_{0.75}$, which are the cases with the highest diameter central cutout, result in the smallest reduction in P_{IGE}/P_{OGE} . The difference between these two cases is negligible. $B_{1.0}O_{0.625}$ and $B_{\infty}O_{0.625}$ fall in between $B_{1.0}O_{0.5}$ and $B_{1.0}O_{0.75}$, showing an attenuation in the reduction of P_{IGE}/P_{OGE} with the increase in the annular opening. Again, the difference between these two cases is minor.

For the annular plates with the larger central cutouts, trends in P_{IGE}/P_{OGE} are not symmetric for positive vs negative h/D . $B_{1.0}O_{0.5}$ is still nearly symmetric, but $B_{1.0}O_{0.75}$, $B_{\infty}O_{0.75}$, and $B_{1.0}O_{0.625}$ actually evince an increase in P_{IGE}/P_{OGE} at $h/D \sim 0.1$ – 0.4 by almost 15% over that of the infinite ground before relaxing toward the OGE condition for larger h/D . Actuator disk theory [18,19] states that in zero freestream, the area of the propeller far wake is half of the propeller disk area when assuming a fully contracted wake. This corresponds to $B = 0.707$, which is smaller than the central cutout of the $O_{0.75}$ cases. This may explain why the attenuation of ground effect for $B_{1.0}O_{0.75}$, $B_{\infty}O_{0.75}$, and $B_{1.0}O_{0.625}$ deviates significantly from trends for the circular plates in the ground effect but not in the ceiling effect. The flow visualization result in the next section will further discuss the trends observed in the force-based experiments.

Figure 5 replots the result shown in Fig. 4a with respect to the blockage ratio B for both ground and ceiling effects, with the selected h/D values. The traditional ground-effect case with a ground plate large enough is noted as ∞ . Again, with the decrement of B , the value of P_{IGE}/P_{OGE} increases, indicating a reduction in the ground-effect benefit. It is also worth noticing that the changes in P_{IGE}/P_{OGE} vs B for the ceiling effect are relatively linear.

The average individual thrust and power coefficients from the same thrust level selected in Fig. 4, normalized by the OGE reference, are plotted in Figs. 6 and 7, respectively. Again, the result for the circular plates is on the left and for the annular plates on the right. As with an infinite ground plate, there is in general an increase in IGE vs OGE thrust and a decrease in IGE vs OGE power. For all cases, a rise in $C_{P\ IGE}/C_{P\ OGE}$ approaching $h/D = 0$ is nearly symmetric, which is to say that the ceiling effect mirrors the ground effect. $C_{T\ IGE}/C_{T\ OGE}$ is also nearly symmetric but only for the circular plates. For the annular plates, the increment in P_{IGE}/P_{OGE} seen in Fig. 4b manifests itself specifically in $C_{T\ IGE}/C_{T\ OGE}$. $B_{1.0}O_{0.625}$ and $B_{1.0}O_{0.75}$ have a considerable drop in $C_{T\ IGE}/C_{T\ OGE}$ at small positive h/D , whereas there

is no such respective change in $C_{P\ IGE}/C_{P\ OGE}$. This drop accounts for the respective loss of propeller performance in Fig. 4b.

As with normalized power at constant thrust, the augmentation of thrust due to the ground or ceiling effect varies for the circular plates inversely with the plate diameter. $B_{1.0}O_{0.0}$ almost exactly follows the curve for $B_{\infty}O_{0.0}$, while $B_{0.5}O_{0.0}$ evinces only a small improvement from OGE. $B_{1.0}O_{0.0}$ also follows the power-ratio curve for $B_{\infty}O_{0.0}$ but not quite as well.

For the annular plates, as expected, plates of smaller opening area have more thrust augmentation and lower power ratio. This is true, whether the plate's outer diameter is one propeller diameter or infinite, $B_{\infty}O_{0.5}$ has noticeably more thrust augmentation than the next smallest cutout, $B_{\infty}O_{0.625}$. But, the two differ little in power ratio.

Recalling Fig. 1, the diameter of the “stagnation streamline” is approximately 50% of the propeller diameter. $B_{1.0}O_{0.5}$ eliminates the region of inward tangential flow while retaining most of the outward tangential flow region. For larger cutouts, the attenuation of thrust argumentation is a result of the radial projected flow on the plate being absent because of the cutout.

The normalized thrust and power coefficients for the APC 14 × 6 propeller for circular ground plates are shown in Fig. 8. Because its γ/D is similar to that of the 17 × 7, so are the results for IGE thrust augmentation and reduction in power absorbed.

A. Flow Visualization

Surface oil flow images are given for circular plates in Fig. 9 and for annular plates in Fig. 10. Most flow patterns can be segregated into two parts, following the observations in Fig. 1. That is, toward the center of the plate, there is the surface projection of a swirling flow. This is bound by what might be termed a projected circular dividing streamline. The surface oil flow visualization is a good representation of the projected mean flow pattern. But, it does not capture the unsteadiness of the flowfield. An unsteady burst from the center swirling flow toward the outer radial flow may occur at the stagnation streamline, where the flow exits the center region. Exterior to this circle, an outward radial flow persists.

For $B_{1.0}O_{0.0}$, Figs. 9a–9c compare $h/D = 0.1, 0.2$, and 1.0 , finding no qualitative difference between 0.1 and 0.2 . But, at $h/D = 1.0$, which from Fig. 4a is toward the edge of any discernible ground effect on force or power, the imprint of the prop wash on the oil film on the plate is weaker. The projected dividing streamline is of larger diameter, and the swirling flow interior to this streamline is less discernible. Because in the physical installation the ground plate is vertical, when the surface-projected flow is weaker, the oil pattern is confounded by the gravity-driven flow of the oil down the plate, visible as a black blotch on the right-hand edge of $B_{1.0}O_{0.0}$ at $h/D = 1.0$. Figure 10 and subsequent flow visualization results then focus on $h/D = 0.1$.

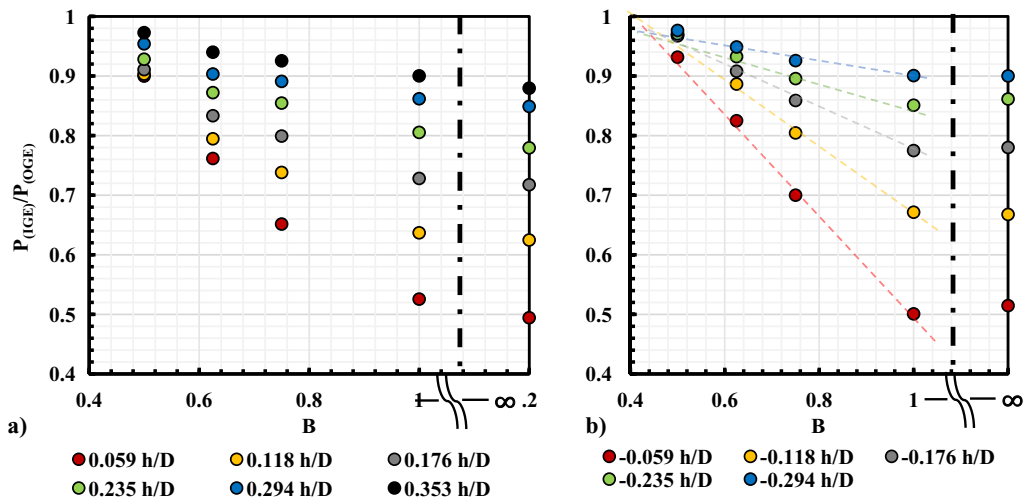


Fig. 5 Normalized power required at constant thrust vs blockage ratio for a) ground effect and b) ceiling effect.

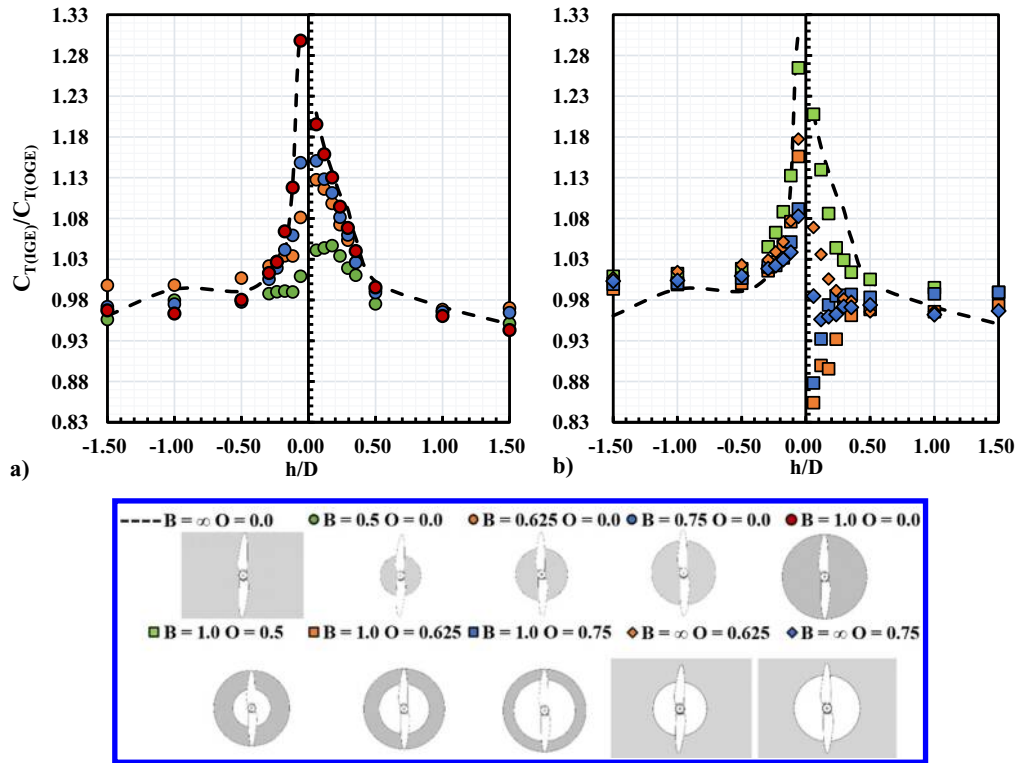


Fig. 6 OGE normalized thrust coefficient of cases in Fig. 4: a) circular and b) annular plates.

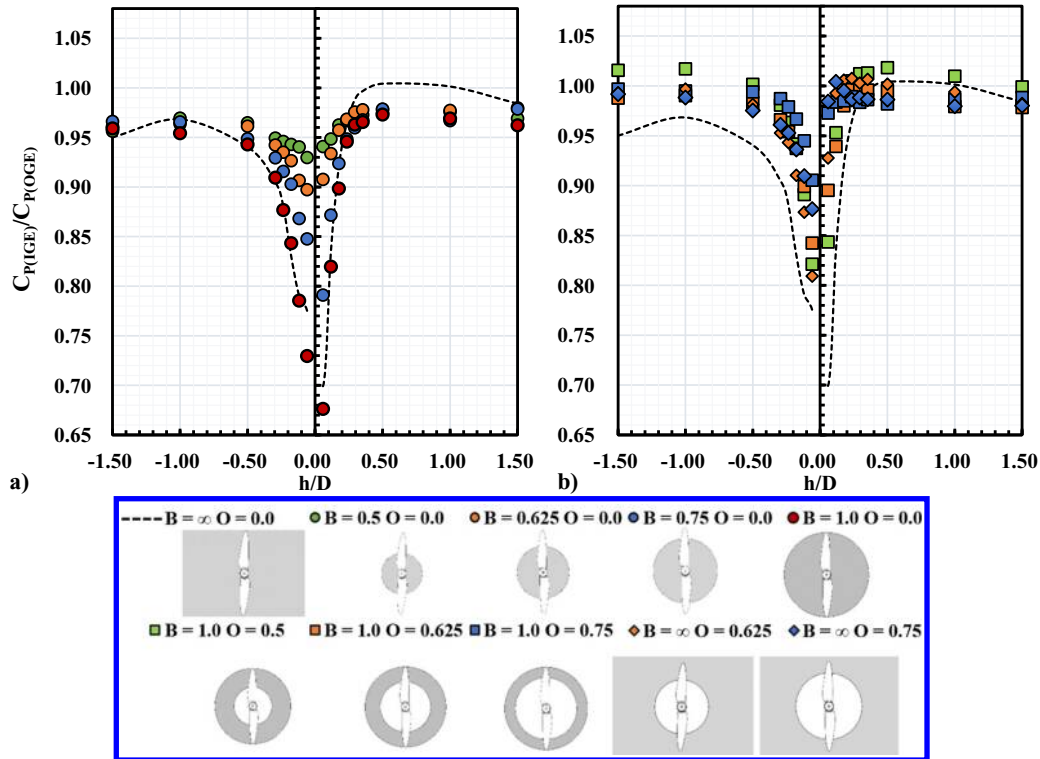


Fig. 7 OGE normalized power coefficient for the cases in Fig. 4: a) circular and b) annular plates.

Given that the IGE thrust augmentation, power reduction, and drop in normalized power at constant thrust of $B_{1.0}O_{0.0}$ follow that of $B_{\infty}O_{0.0}$, it is reassuring that the two surface-projected flowfields on the respective ground plates in Figs. 9c and 9d are similar. Reducing the circular plate diameter results in a smaller radial flow region, as seen in the $B_{0.75}O_{0.0}$ case. This corresponds to the reduction of ground-effect benefits in thrust and power. Further decrement of

the circular ground-plate radius leads to the elimination of the radial flow region, where the projected dividing streamline nears the edge of the circular ground plate. The loss of a radial outflow region tracks with nearly complete loss of ground-effect benefits in thrust and power for $B_{0.5}O_{0.0}$. This is consistent with the theory of Betz [1], that is, considering the ground effect as a “cushion effect,” the flow pressure at the radial flow region along the near surface of the ground

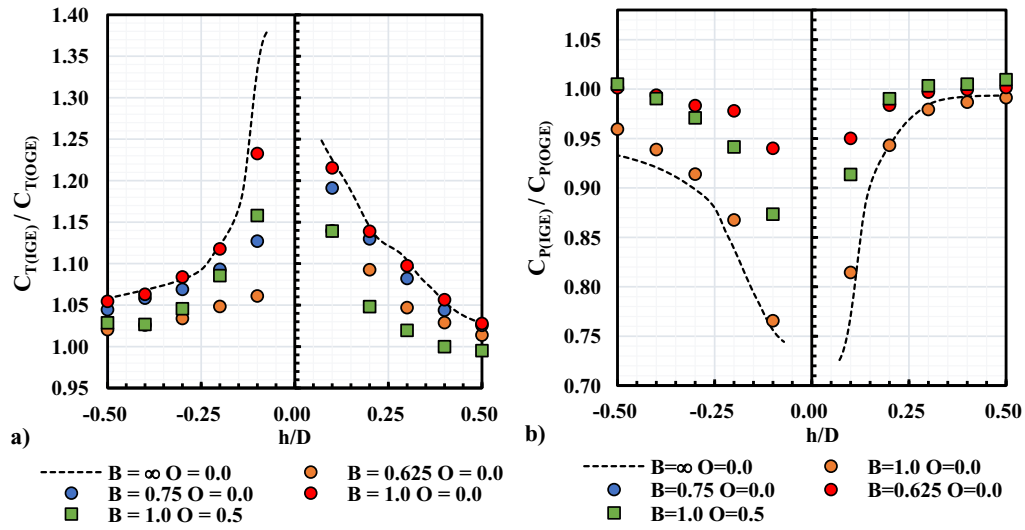


Fig. 8 Normalized a) thrust coefficient and b) power coefficient for the APC 14 $\times$ 6 propeller.

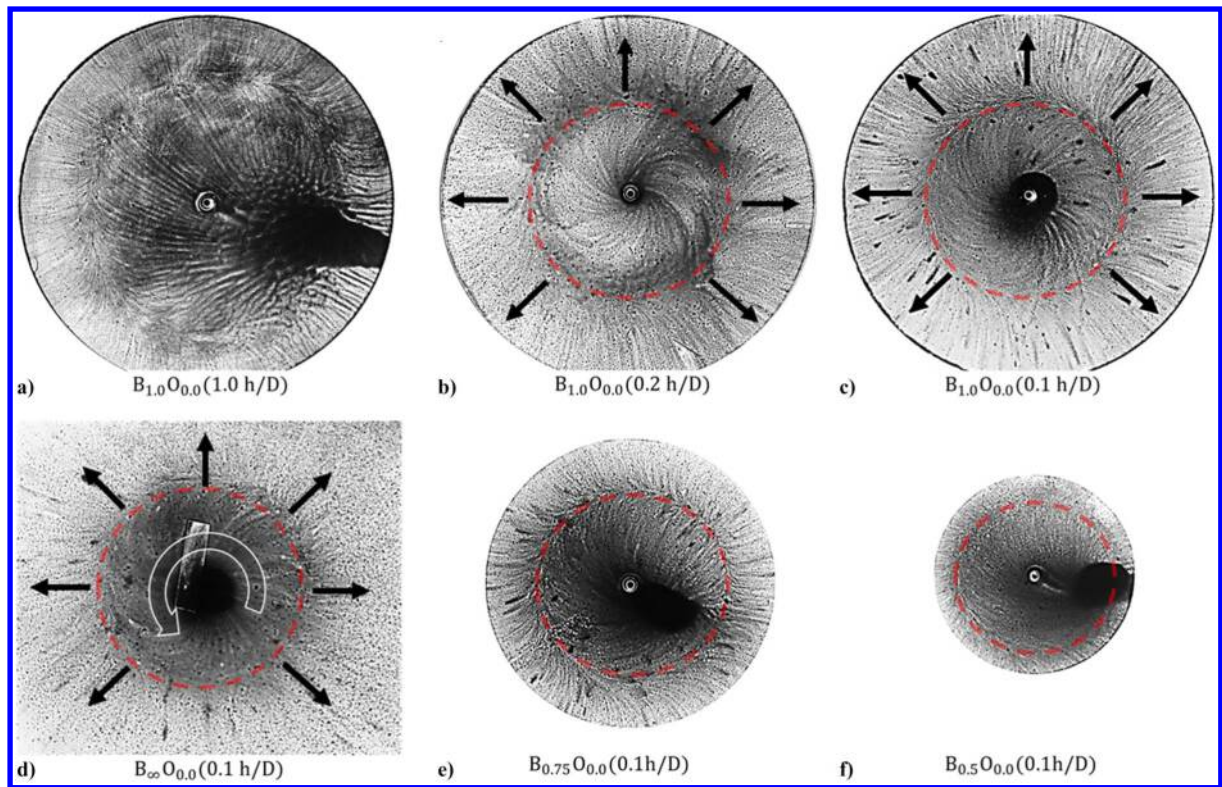


Fig. 9 Surface oil flow: a) $B_{1.0}O_{0.0}(1.0h/D)$, b) $B_{1.0}O_{0.0}(0.2h/D)$, c) $B_{1.0}O_{0.0}(0.1h/D)$, d) $B_{\infty}O_{0.0}$, e) $B_{0.75}O_{0.0}$, and f) $B_{0.5}O_{0.0}$.

plate increases, resulting in thrust augmentation. Thus, the absence of said flow region may plausibly account for the loss of ground-effect benefits.

For the annular plates, a clear projected dividing streamline occurs in $B_{1.0}O_{0.5}$, resembling a similarly projected streamline for $B_{1.0}O_{0.0}$. This is consistent with the similarity in thrust and power IGE results for the two plates. As the circular cutout increases, the projected dividing streamline expands to a larger diameter and evidently weakens. That is, the near-surface flow visualization for an annular plate with a large cutout, at $h/D = 0.1$, comes to resemble that for a circular plate at $h/D = 1.0$.

We next turn to flow visualization by smoke injection for circular plates (Fig. 11) and annular plates (Fig. 12), again at $0.1h/D$. The effect of radial location of the injection point is considered by comparing injection inboard ($0.5D$) and outboard ($0.8D$) of the

projected dividing surface streamline. The yellow dots in Figs. 12a and 12b mark the approximate location of the proposed dividing surface streamline for each respective case, from Fig. 10.

In Fig. 11, when the smoke injection is inboard of the proposed dividing surface streamline, smoke coalesces in a circular area of radial extent interior to said line (Fig. 11a). Injecting near the location of the streamline results in part of the smoke moving inboard and part radially outboard. Injecting the outboard of the streamline (Fig. 11c) sends all the smoke radially outward and away from the ground plate. For $B_{0.5}O_{0.0}$, the smoke just converts axially past the perimeter of the plate. The larger the plate diameter in Fig. 11 is, the stronger is the outboard radial flow. For $B_{1.0}O_{0.0}$, the radial outflow resembles the canonical pattern for an infinite ground plate.

Figure 12 considers the annulus cases $B_{1.0}O_{0.5}$, $B_{1.0}O_{0.75}$, $B_{1.0}O_{0.625}$, and $B_{\infty}O_{0.625}$. At $0.5D$ injection point, stagnation point

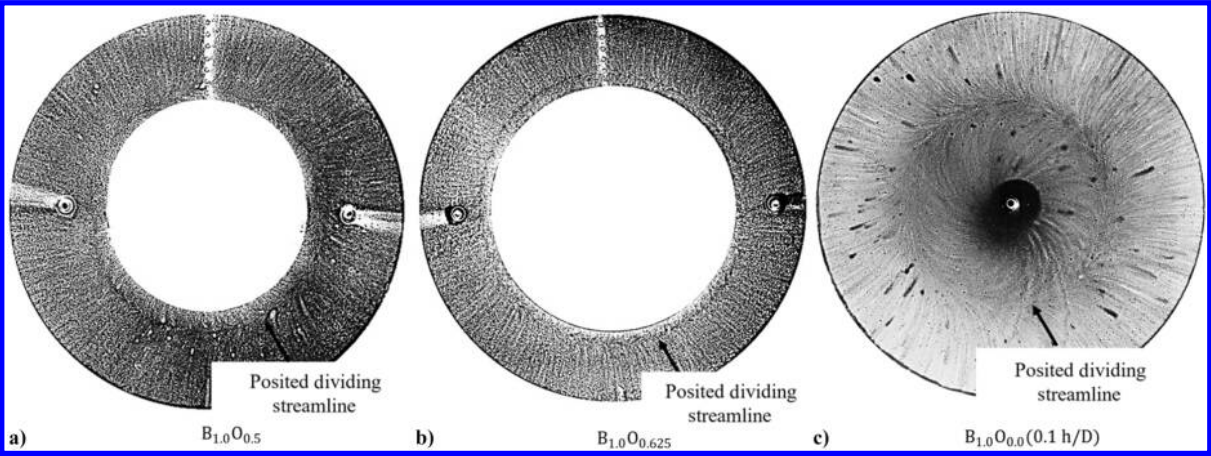


Fig. 10 Surface oil flow visualization for $B_{1.0}O_{0.5}$ (left), $B_{1.0}O_{0.625}$ (middle), and $B_{1.0}O_{0.0}$ (right); $h/D = 0.1$.

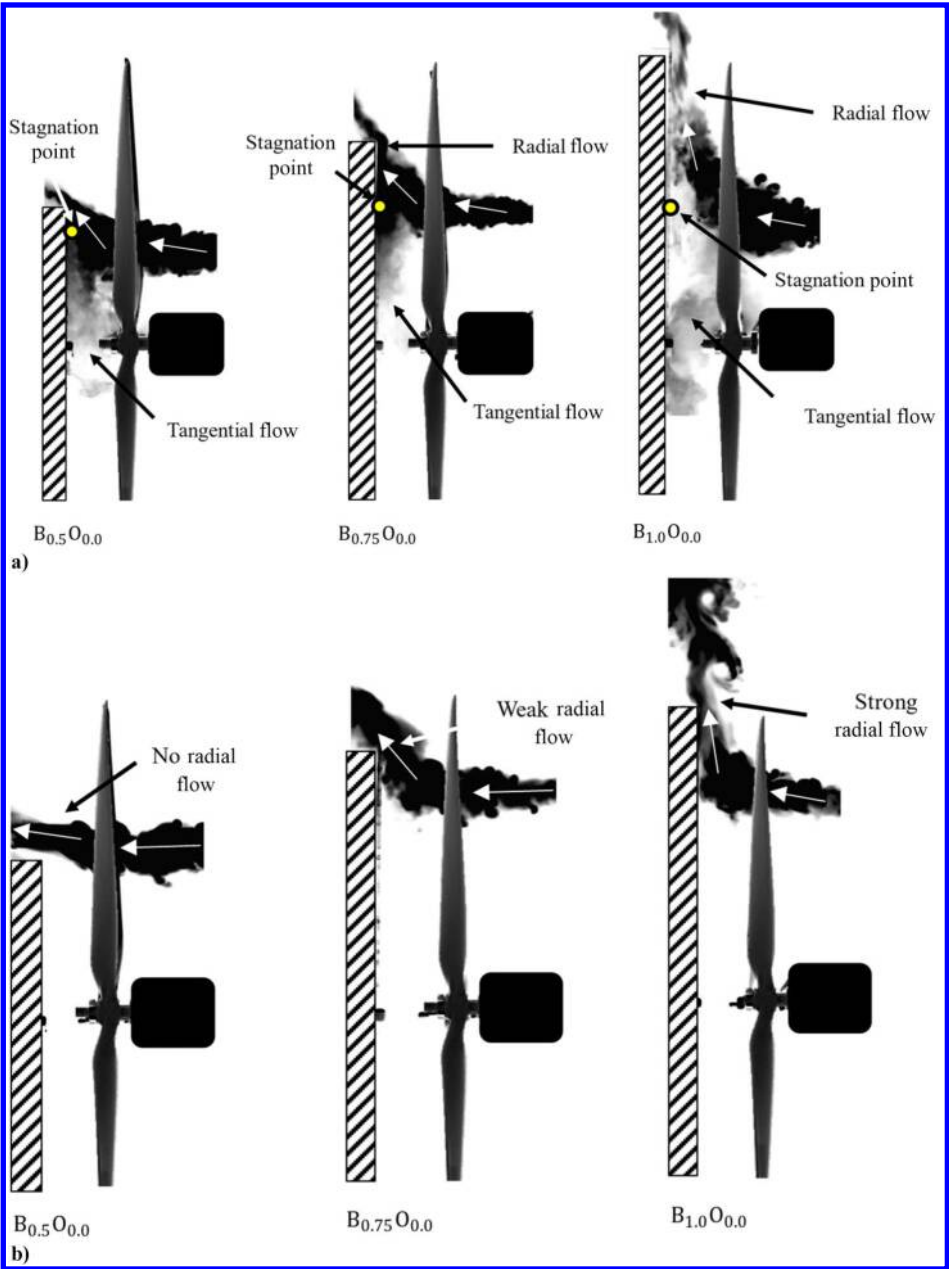


Fig. 11 $B_{0.5}O_{0.0}$, $B_{0.75}O_{0.0}$, and $B_{1.0}O_{0.0}$; $h/D = 0.1$ (ground effect); smoke injection at a) $0.5D$ and b) $0.8D$.

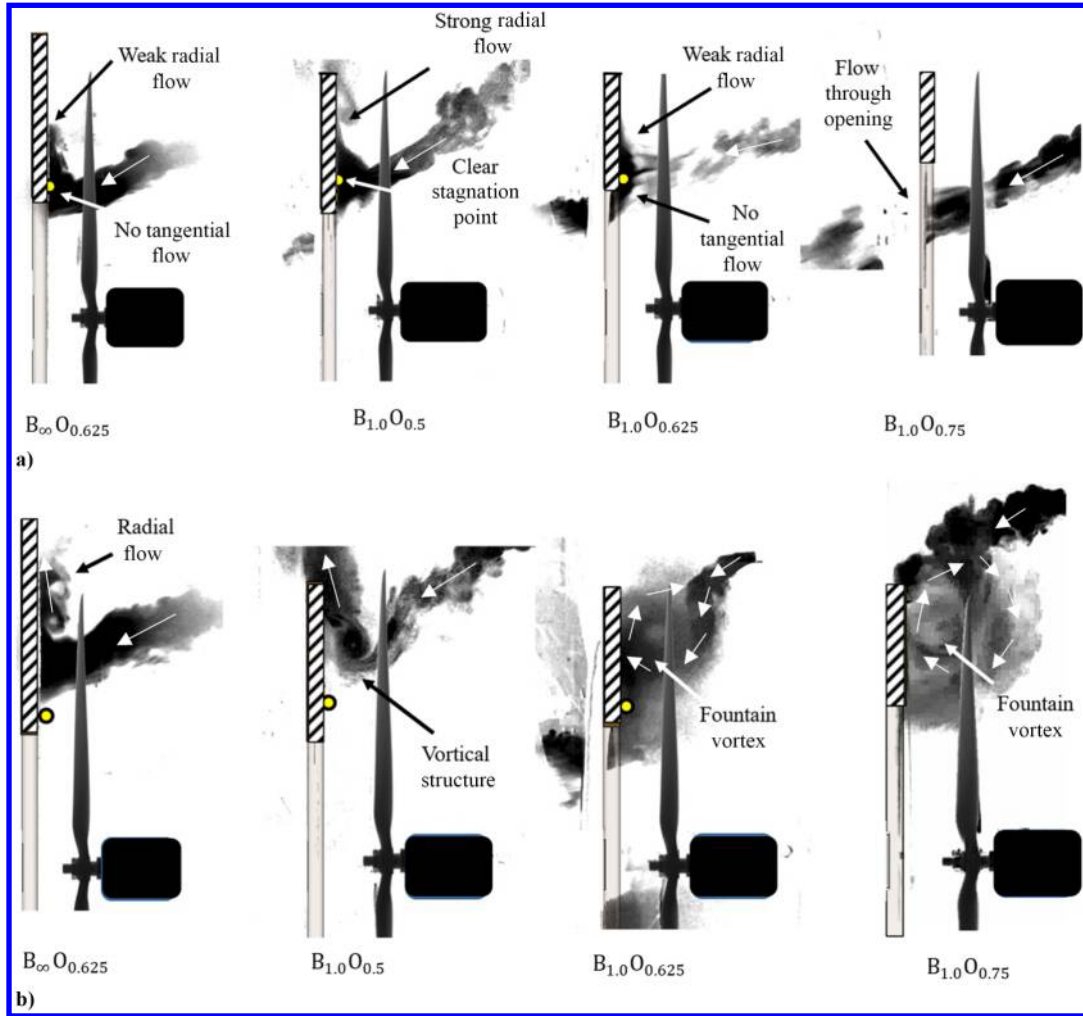


Fig. 12 $B_{1.0}O_{0.5}$, $B_{1.0}O_{0.625}$, $B_{\infty}O_{0.625}$, and $B_{1.0}O_{0.75}$; $h/D = 0.1$ (ground effect); injection at a) $0.5D$ and b) $0.8D$.

for the $O_{0.5}$ case posited some distance from the edge of the opening, persevering both tangential flow and radial flow. When increasing the annulus cutout, the stagnation point occurs at the edge of the opening for both $O_{0.625}$ cases, resulting in the elimination of the tangential flow region. A qualitative difference can be seen for the $O_{0.75}$ case as the flow goes through the center opening directly.

The $0.8D$ injection point shows for all cases in Fig. 12a “fountain vortex” with clockwise rotation near the tip of the propeller. It is present whether or not there is a projected dividing surface streamline on the plate. This vortex has been associated with the decrease in thrust coefficient as the effect blade angle of attack is reduced at the propeller tip [11,12,28]. Evidently, the annulus cutout affects both radial and tangential flow regions. The smoke injection cannot rigorously track vorticity, but there is a qualitative signature of a vortical structure in particular for $B_{1.0}O_{0.5}$, marked in Fig. 12b.

B. Superposition of IGE Effects

To see whether P_{IGE}/P_{OGE} for the annular and circular plates superimpose, we define a normalized power deficit:

$$P_{def} = 1 - \frac{P_{IGE}}{P_{OGE}} \quad (1)$$

P_{def} is the “height” of the dip in the curve in Fig. 4 below 1.0 and is calculated for each individual case in Table 1, for each h/D . Then, P_{def} for $B_{0.75}O_{0.0}$ and $B_{\infty}O_{0.75}$ are summed and compared with P_{def} for $B_{\infty}O_0$, as it was inserting the $0.75D$ circular plate into the infinite ground with $0.75D$ annulus. Similarly, P_{def} for $B_{0.75}O_{0.0}$ and $B_{1.0}O_{0.75}$ are summed and compared with P_{def} for $B_{1.0}O_0$. This is done in Fig. 13 for every case, where the circular plate complements

the respective annular plate, the latter either of $1.0D$ outer diameter or infinite, resulting in a circular plate of, respectively, $1.0D$ or infinite diameter.

All curves in Fig. 13 overlap for the ceiling effect ($h/D < 0$). Recall that the IGE thrust and power results for the ceiling effect all follow a monotonic trend toward $h/D \rightarrow 0$, whereas on the ground-effect side the agreement is not as good. It is appealing to speculate that the ceiling effect can be represented by a function of the plate disk area for any type of plate geometry.

For the ground-effect side ($h/D > 0$), $B_{0.5}O_{0.0} + B_{1.0}O_{0.5}$ closely resembles $B_{1.0}O_{0.0}$. $B_{0.625}O_{0.0} + B_{\infty}O_{0.625}$ and $B_{0.75}O_{0.0} + B_{\infty}O_{0.75}$ nearly follow $B_{1.0}O_{0.0}$, which is equivalent to $B_{\infty}O_{0.0}$, as discussed previously. However, a deviation occurs for $B_{0.625}O_{0.0} + B_{1.0}O_{0.625}$ and $B_{0.75}O_{0.0} + B_{1.0}O_{0.75}$ around $h/D = 0.3$, which is a result of the abnormal behavior seen in Fig. 4 at small positive h/D . This abnormality is predicated toward the presence of the recirculation region and the fountain vortex in these cases, as seen in Fig. 11b.

C. Pressure Measurements

Figure 14 shows normalized pressure along a radial line of the ground plate for $B_{1.0}O_{0.0}$ at different h/D . The pressure is normalized in two ways. The first is by the ratio of the force measured on the plate (via the load cell simultaneously) at the same h/D , divided by plate area (Fig. 14a):

$$P_{d, norm} = \frac{P_x}{4D_p/\pi d_p^2} \quad (2)$$

The second is by the propeller disk loading (Fig. 14b), which conveniently also has units of pressure. The disk loading itself

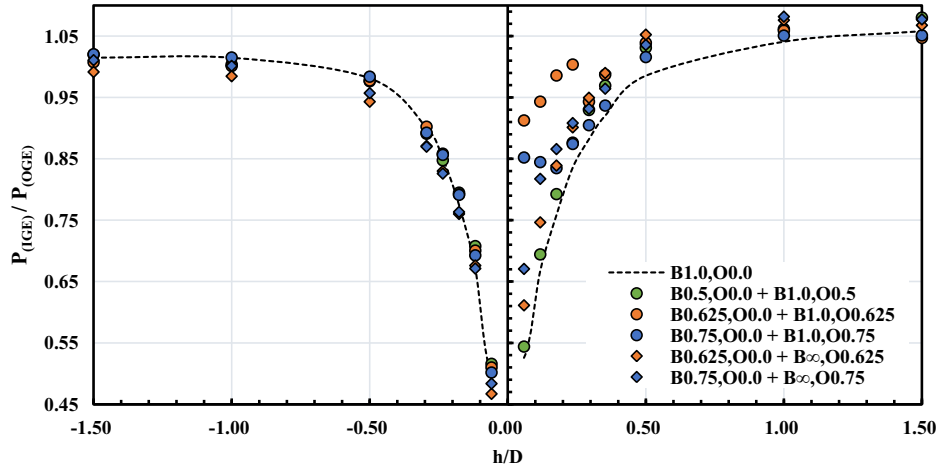


Fig. 13 Power required at constant thrust, summing complementary annular and round plate cases.

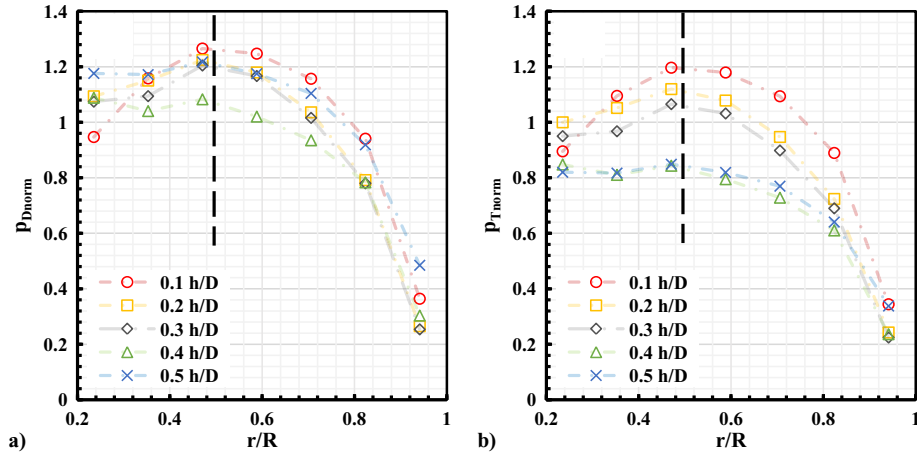


Fig. 14 $B_{1.0}O_{0.0}$ plate pressure normalized by a) plate force and b) propeller disk loading.

varies with attained thrust, and hence with h/D or the ground-plate type:

$$p_{T_{\text{norm}}} = \frac{p_x}{4T_{\text{IGE}}/\pi d^2} \quad (3)$$

Regardless of normalization, the peak pressure is near the $0.5r/R$ radial station, which is also in the vicinity of the surface-projected dividing streamline. Pressure approaches zero at the plate edge, $r/R = 1.0$. In the normalization by plate force loading, there is no clear trend in pressure variation with h/D . Evidently, as h/D increases, the shape of the pressure distribution does not change, and there are no discernible trends in magnitude in this normalization because the integral of pressure over the area ought by definition to be the total force, assuming ambient pressure on the back face of the ground plate. But, in the normalization by disk loading, peak pressure falls as h/D increases. This normalization will be used in the subsequent discussion.

Pressure distribution for $B_{1.0}O_{0.0}$ in ceiling effect, normalized by propeller disk loading, is shown in Fig. 15. At $h/D = -0.1$, normalized pressure remains around 0.7 at the center of the plate and increases significantly toward the plate edge. This is the only h/D value at which there is a strong relationship of pressure with r/R , as at all other h/D the curve is flatter with r/R . As h/D increases, the overall pressure curve falls, tending toward zero as a result of the reduction in ceiling effect.

Pressure data normalized by propeller disk loading for $h/D = 0.2$ and $h/D = -0.2$ are shown in Fig. 16 for the full roster of cases. For the three smaller circular plates ($B_{1.0}O_{0.75}$, $B_{1.0}O_{0.625}$, and $B_{1.0}O_{0.5}$), the pressure is lower than for $B_{1.0}O_{0.0}$, but there is a significant drop at

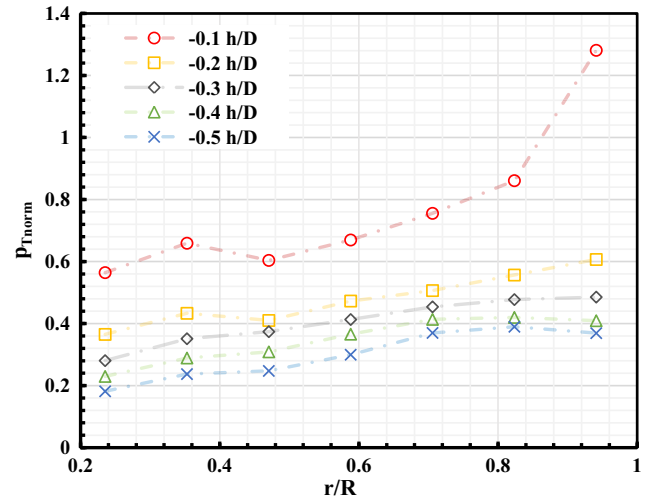


Fig. 15 $B_{1.0}O_{0.0}$ pressure in ceiling effect, normalized by propeller disk loading.

the outer edge of the plate for all three cases. For $B_{1.0}O_{0.5}$ and $B_{1.0}O_{0.625}$, the pressure distribution follows that of $B_{1.0}O_{0.0}$ at $r/R > 0.6$ and $r/R > 0.75$, respectively. For $B_{1.0}O_{0.75}$, speculatively due to the fountain vortex seen in Fig. 12b, the pressure is lower than for $B_{1.0}O_{0.0}$ at all r/R , which would further reduce the propeller IGE thrust. This agrees with the result seen in Fig. 6b. Both phenomena led to the result shown in Fig. 13, where the partial ground-effect

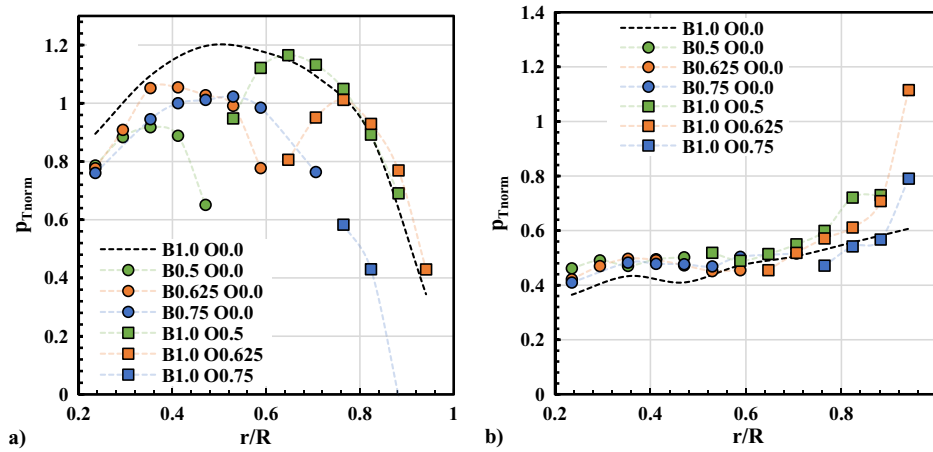


Fig. 16 Normalized surface pressure vs r/R at a) $h/D = 0.2$ and b) $h/D = -0.2$.

results for the circular plate and annular plate cannot be added together to accord with that of $B_{1.0}O_{0.0}$.

The result for $h/D < 0$ (ceiling effect) is shown in Fig. 16b. Overlap of normalized pressure, whether for circular plates or annuli, is tighter than in the ground-effect case. This corroborates the phenomenon in Fig. 13, where the thrust in the ceiling effect is addable based on the plate area.

The measured force acting on each circular ground plate can be compared with the area integral of the pressure on the windward side, assuming that the pressure on the leeward side is ambient and that the single azimuth along which the taps are located is representative of the entire plate, which can be expressed as

$$D_{p(x)} = \int_{r_o}^{r_b} p(x) 2\pi r dr \quad (4)$$

The ratio of measured to calculated force on the circular plates is given in Fig. 17a. For $|h/D| = 0.2$ or less, this ratio is between 0.8 and 1.2. Agreement gets worse at larger distance from the plate. In Fig. 17b, force on the plate is normalized by the propeller thrust in OGE. On the ceiling-effect side, this ratio tends to zero, for $h/D \rightarrow -0.5$, meaning that the role of the plate becomes negligible once the plate is one-half of the propeller diameter away, falling to ~ 0.1 even for the largest plate, $B_{1.0}O_{0.0}$. On the ground-effect side, for $B_{1.0}O_{0.0}$, the force on the plate is still some 70% of the propeller thrust at $h/D = 0.5$. Force on the plate actually exceeds the OGE propeller thrust because the IGE thrust is 20% larger than OGE (Fig. 6a). For smaller plates, the ratio of plate force to propeller thrust is of course lower than for larger plates, for any h/D .

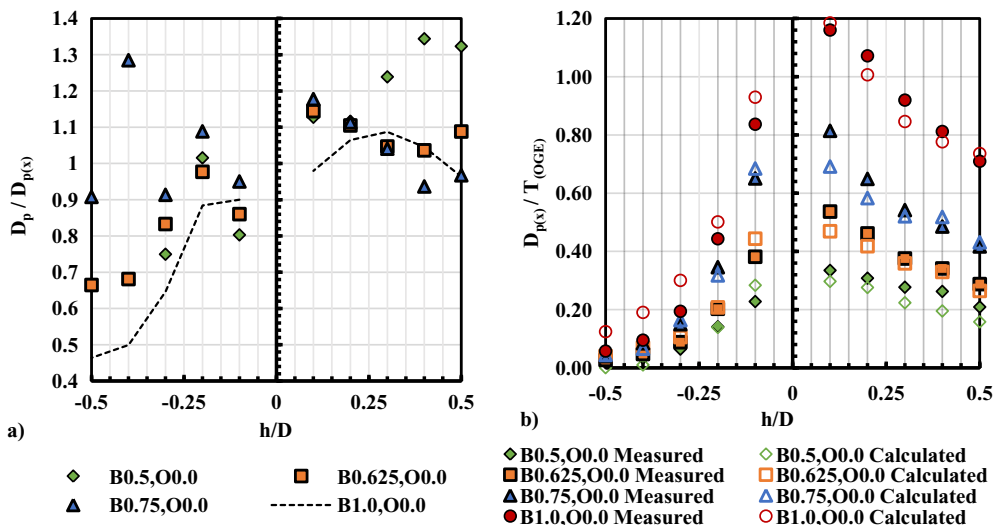


Fig. 17 Ratio of a) measured force on plate to pressure integration and b) pressure integration to OGE propeller thrust vs h/D .

As the propeller wake partially passes through the interior of the annular plates, a direct consideration of force on the annular plates is perhaps not instructive. Instead, the annular plates are summed with their respective “interior” circular counterparts. Directly measured respective forces, and force inferred from area integration of pressure, are then again divided by the OGE propeller thrust. The result is shown in Fig. 18.

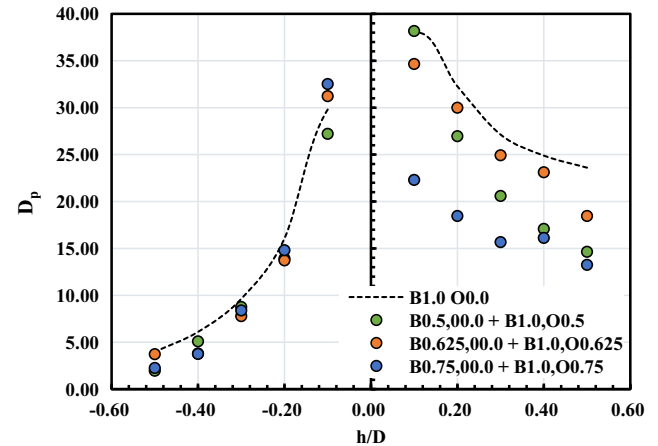


Fig. 18 Force on plates from pressure data, summing complementary annular and round cases.

As with the summation of P_{IGE}/P_{OGE} in Fig. 13, for $h/D > 0$ (ground effect), the various cases do not collapse to one curve; the normalized force on the summed cases is consistently lower than that on the reference case, $B_{1.0}O_{0.0}$. But, for $h/D < 0$, the various cases closely follow the reference trend. Again, this indicates that the ceiling effect depends on the plate area and not the plate geometry.

D. Net Force Summation

To further understand the connection between propeller thrust and force acting on the circular ground plate, two additional propellers, APC 17 × 10 and 17 × 12, were tested. The different γ/D propellers have starkly different thrust, both in OGE and at various h/D [12], and larger h/D results in diminished IGE benefit. Again, the propeller thrust and the force on the circular plate are measured simultaneously at different h/D . The net force summation is

$$F_{\text{net}} = T + D_p \quad (5)$$

The force acting on the circular ground plate will be negative, because in ground effect it pushes on the plate. The net force can be regarded as a residual thrust after accounting for blockage by the plate, if say the propeller were affixed on an aircraft and the plate were a fuselage behind the propeller. The net force, normalized by the propeller thrust in OGE, is shown in Fig. 19.

Despite diminished IGE benefit for large γ/D , normalized net force is nearly independent of γ/D and increases more or less monotonically within $0 < h/D < 1$, leveling off for $h/D > 1$. There is a clear dependency on the ratio of the plate to propeller diameter, with smaller normalized net force for the larger plates. As the increase in D_p for the larger plates overwhelms the stronger IGE thrust benefit. For the largest plate, $B_{1.0}O_{0.0}$, the normalized

net force approaches 0 for $h/D = 0.05$. As intuitively the result should also be 0 for $B_{\infty}O_{0.0}$, this is further evidence that a ground plate of the same diameter as the propeller is already essentially infinite.

On the ceiling-effect side, the normalized net force tends to 1.0, for $h/D < -1$, for all of the plates. Momentum theory says that the stream-tube diameter far upstream of the propeller is infinite, and so it intuitively holds that a finite diameter plate would have a negligible effect at sufficiently large h/D . The result in Fig. 19 implies that this “sufficiency” is $h/D < -1$. But, on the ground-effect side, even $h/D = 1.5$ is evidently not enough for the normalized net force to reach 1.0 or for results for the different plates to converge to the same value. Indeed, this value must be less than 1, for otherwise it would have been the case that the placement of a blockage in the propeller wake somehow produces a net thrust.

E. Prediction Model Adapting Circular Ground Plate

An empirical curve fit, not necessarily contingent on physical intuition, can be built from the thrust and power measurements and tested against a case outside of those from which the model was tuned. From data for the 17 × 7 propeller, taking the power-required deficit P_{def} vs plate diameter ratio B for different h/D , Eq. (1) can be modified as

$$\frac{P_{B \text{ IGE}}}{P_{\text{OGE}}} = \begin{cases} 1 - (P_{\text{def}})B & (0 \leq \frac{h}{D} \leq 0.5) \\ 1 - (P_{\text{def}})B^{\sqrt{2}} & (-0.5 \leq \frac{h}{D} \leq 0) \end{cases} \quad (6)$$

Equation (6) is only applicable to circular plates with $B = 1$ or less. For any ground plate with $B > 1.0$, the measured result will follow

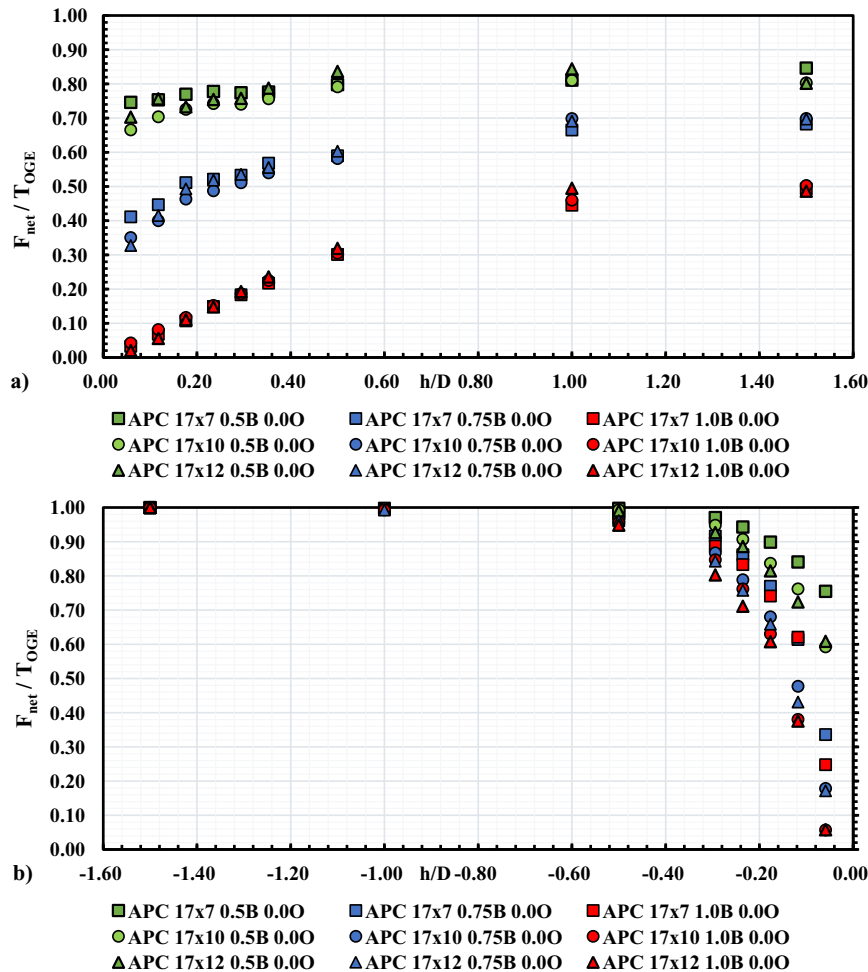


Fig. 19 Normalized net force summation for APC 17 × 7, 17 × 10, and 17 × 12 propellers in a) ground effect and b) ceiling effect.

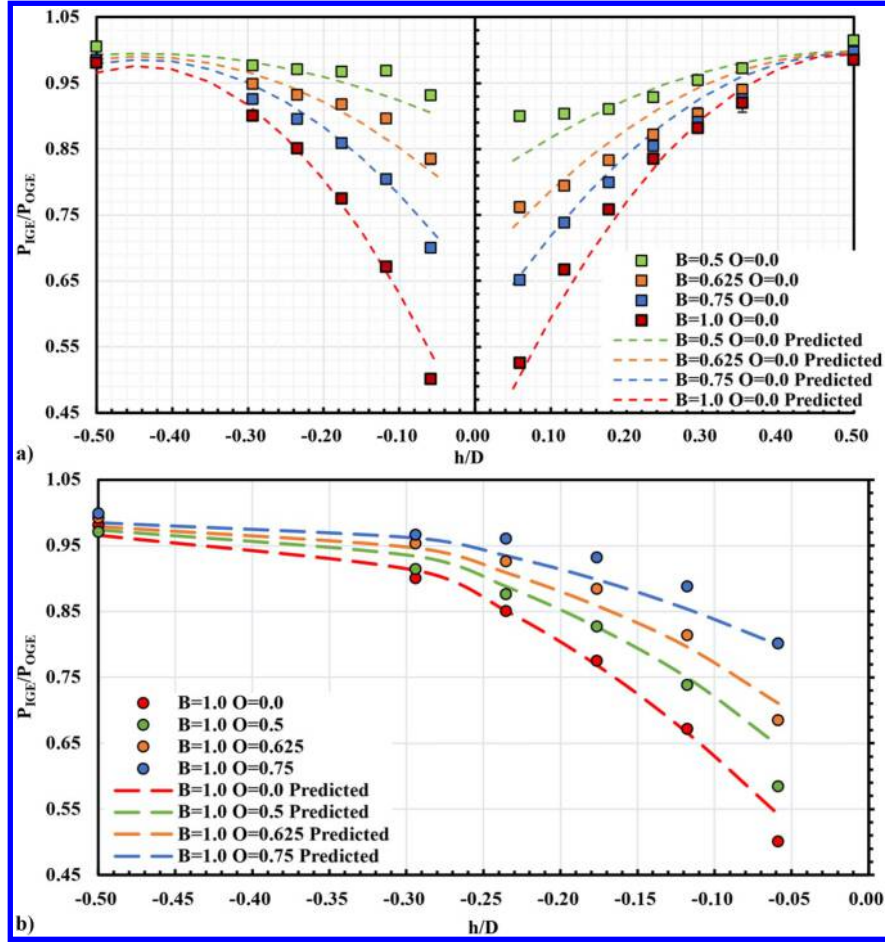


Fig. 20 Predicted vs measured $P_{B\text{ IGE}}/P_{\text{OGE}}$ for APC 17×7 propeller with a) circular ground plate and b) annular ground plate.

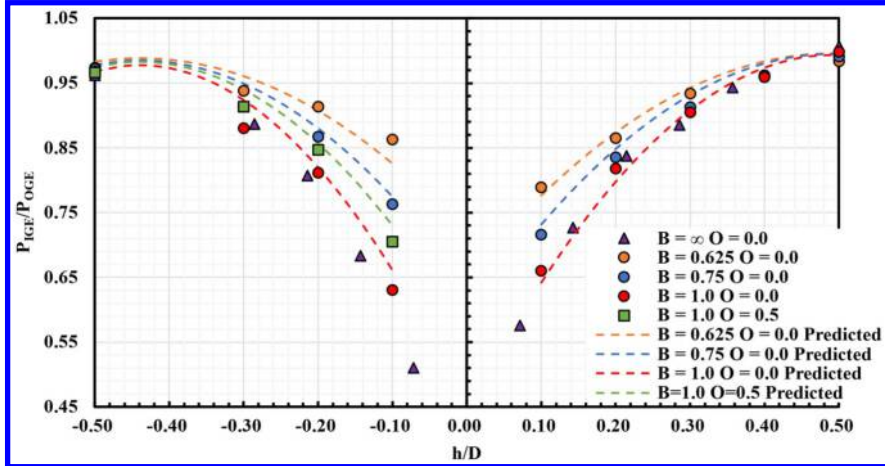


Fig. 21 Predicted vs measured $P_{B\text{ IGE}}/P_{\text{OGE}}$ for APC 14×6 propeller.

$B = 1.0$, as seen in Fig. 4. Recall from Fig. 18 that the ceiling effect can be predicted as a function of the plate area. From the pressure data in Fig. 16, it is fair to assume that the pressure distribution on the ground plate in the ceiling effect is uniform radially. An alternative method to predict the ceiling effect for arbitrary plate shape is

$$\frac{P_{\text{Ce IGE}}}{P_{\text{OGE}}} = 1 - \frac{\int_0^B 2\pi r dr}{\pi r^2} * P_{\text{def}(B1.0\text{ } O0.0)} \left(-0.5 \leq \frac{h}{D} \leq 0 \right) \quad (7)$$

The models are compared with the experimental data from which they are sourced (17×7 propeller) in Fig. 20 and with data from another APC thin electric series propeller, the 14×6 in Fig. 21.

Equation (3) is used for the prediction of circular ground plates, while Eq. (4) is used for the prediction of annular ground plates. Neither model is asserted to be an optimization of any kind but instead merely illustrates that a unifying trend can be discerned. For $B_{0.5}O_{0.0}$, the fit is not broadly successful, but for all of the larger circular plates scatter in both ground and ceiling effect is small, with an overall error of less than 6%.

IV. Conclusions

Propeller ground effect and ceiling effect in hover were measured experimentally for circular and annulus-shaped ground plates of various sizes, with the aim of assessing the effect of plate size and

shape on propeller thrust and power as a function of proximity to the plate. Smoke and surface flow visualization revealed a region of circumferential flow toward the inner part of the plate, with an outer radial flow region. The strength of the radial flow was found to correlate positively with increase in thrust and decrease in power required associated with ground proximity. Visualization revealed a flowfield structure sometimes dubbed a fountain vortex. In cases where the fountain vortex was qualitatively strong, the ground-effect benefits on thrust and power were found to be negligible.

For circular plates of the same diameter as the propeller, the ground effect and ceiling effect followed those of an infinite plate, suggesting that any further increase in plate diameter is not necessary for a ground-interaction study, at least in terms of force/power measurement at the propeller shaft. Reduction in the circular plate diameter results in a reduction of both thrust augmentation and power savings, with almost negligible changes in the propeller performance in proximity to a circular plate with a diameter ratio of 0.5, where projected radial flow no longer presents on the ground plate. The force summation of the ground and propeller can be considered as a function of the normalized ground-plate size, as the effect of the propeller geometry is negligible in this scenario.

For annular ground plates with a small inner diameter, a strong outward radial flow was found, which results in similar changes in the thrust coefficient when compared to infinite ground. With greater inner diameter of the annulus, the benefit of the ground effect reduces. Further increment of the annulus inner diameter results in a loss in the propeller thrust generation, which attributed to the formation of a fountain vortex.

The ceiling-effect benefits, represented by P_{IGE}/P_{OGE} , are found to be additive; IGE thrust augmentation for a small circular plate, plus that of an annular plate of inner diameter equal to the outer diameter of the former, sums to the respective IGE effect of a round plate with the same outer diameter as the annulus. This suggests that the partial ceiling effect depends just on the ground-plate area and not on its shape. Such summation of partial ground effect, on the other hand, is not successful, again perhaps due to the presence of the fountain vortex in the annulus cases. An algebraic model developed for infinite ground was modified for circular plates, showing normalized power required in ground-effect and ceiling-effect prediction within 6% error.

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